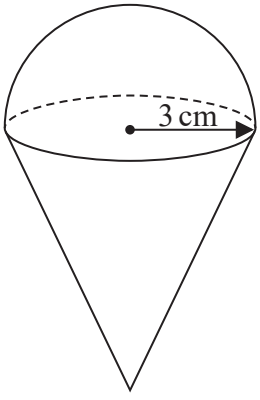
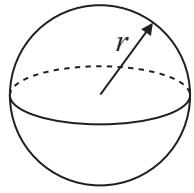
19 The diagram shows a solid made by joining a solid hemisphere to a solid circular cone.

The centre of the plane face of the cone coincides with the centre of the plane face of the hemisphere.



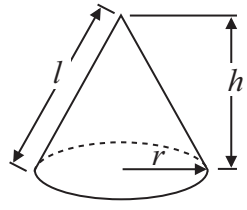
Surface area of sphere = $4\pi r^2$

Volume of sphere = $\frac{4}{3}\pi r^3$



Curved surface area of cone = πrl

Volume of cone = $\frac{1}{3}\pi r^2 h$



The radius of the hemisphere is 3 cm.
The radius of the base of the cone is 3 cm.

The volume of the solid is 30π cm^3

Work out the total surface area of the solid.
Give your answer as a multiple of π.

Hemisphere volume = ½ × (4/3)π × 3^3 = 18π

1 mark
Split the total volume.

Cone volume = 30π − 18π = 12π

(1/3)π × 3^2 × h = 12π, so 3h = 12, h = 4

1 mark
Find cone height 4.

Slant height l = √(3^2 + 4^2) = 5 cm

1 mark
Pythagoras gives slant 5.

Exposed area = 2π × 3^2 + π × 3 × 5

1 mark
Add the two curved areas.

= 18π + 15π = 33π cm^2

1 mark
33π cm^2.

.....cm^2

(Total for Question 19 is 5 marks)